

# Homework 1

## Sector Deep Dive

*Big Data Analytics Module – Prof. Giovanni Morana*

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**Delivery:** Jupyter Notebook (.ipynb)

**Deadline:** May 15th

**Grading:** All steps must be completed to access Homework 2

## Your Sector Assignment

Find your assigned sector in the table below using the last digit of your Student ID.

| Student ID (ending with...) | Sector                 |
|-----------------------------|------------------------|
| 1                           | Healthcare             |
| 2                           | Financial Services     |
| 3                           | Consumer Cyclical      |
| 4                           | Consumer Defensive     |
| 5                           | Communication Services |
| 6                           | Energy                 |
| 7                           | Industrials            |
| 8                           | Basic Materials        |
| 9                           | Real Estate            |
| 0                           | Utilities              |

## How to Document Your Work

*You do not need a markdown cell after every chart. What is required is that your code is commented throughout: explain what each block does, and whenever you make a choice (which metric to use, which industries to keep, which quadrant to focus on) write a short comment explaining why. The only mandatory written section is the Final Summary at the end.*

## Step 0 — Setup and Sector Filter

Load the dataset and apply the same cleaning seen in class: keep only active companies, remove ETFs and funds. Then filter to your assigned sector.

## Step 1 — Market Capitalization Distribution

Plot a **histogram** of **market\_cap** for your sector. Add two vertical lines: one for the mean, one for the median. Both must be annotated with their value.

## Step 2 — Outlier Detection

Plot a **boxplot** of **market\_cap**. If needed, remove outliers.

Use the cleaned dataset from this point forward.

## Step 3 — Industry Comparison: Four Metrics

Create a **single figure with 4 subplots**. Each subplot is a horizontal bar chart showing the **mean value per industry** for one of the following metrics:

- market\_cap
- total\_revenue
- profit\_margins
- beta

## Step 4 — Industry Dispersion

Create **4 boxplots**, one per metric (**market\_cap**, **total\_revenue**, **profit\_margins**, **beta**). Each boxplot must show the full distribution broken down **by industry**, not just the mean.

## Step 5 — Focus Industry Selection

Based on Steps 3 and 4, **select 3 or 4 industries** to carry forward. Filter the dataset to those industries only.

In a comment or markdown cell, justify your choice for each industry you keep and each one you drop. The justification must reference specific values or patterns you observed in the charts, not general opinions about the industry.

## Step 6 — Bubble Chart

Using the filtered dataset from Step 5, create a bubble chart. You decide:

- what goes on the x-axis
- what goes on the y-axis
- what determines bubble size
- what determines bubble color

Document these four choices in a comment before the code, and explain what each encoding is meant to reveal.

Divide the chart into 4 quadrants. Each quadrant represents a combination of the two axis values: for example, if you chose beta on the x-axis and market\_cap on the y-axis, the four quadrants will represent companies that are simultaneously high or low on each dimension (high beta and high cap, low beta and high cap, and so on). Label each quadrant with the combination it represents..

## Step 7 — Zoom into One Quadrant

Choose one quadrant from Step 6 and justify your choice in a comment. Filter to the companies in that quadrant. If there are more than 10, keep the top 10 by a dimension you prefer.

Use the yfinance example seen in class to download the last 3 months of daily closing prices. Normalize all series to 1 on the first available trading day and plot them on a single chart with a legend identifying each company.

**In your comments, reflect on what you observe in the price chart in relation to the current geopolitical situation, in particular the tensions in the Strait of Hormuz.**

Were the companies in this quadrant visibly affected?

Do you see a sharp drop, a gradual decline, or no significant impact?

Is there any sign of recovery?

## Step 8 — Compare with an Excluded Quadrant

Pick one of the three remaining quadrants from Step 6.

Apply the same procedure as Step 7: take the top 10 companies by the same dimension, download the last 3 months of prices, normalize, and plot on a single chart.

**In your comments, compare the behavior of this group with the one in Step 7.**

Did these companies react differently to the same geopolitical context?

Were they more resilient, more exposed, or did they follow a similar pattern?

Comment on whether the quadrant logic, i.e. the combination of metrics you used to define it, helps explain any difference (if any) in behavior.

## Final Summary

Write a summary of your analysis in **no more than 400–500 words** in a markdown cell at the end of the notebook. Cover:

- the overall structure of your sector: size, profitability, and risk
- why you selected the industries you did
- what the bubble chart revealed about your focus universe
- how the two groups of companies performed over the last 3 months, and whether any difference, or lack of difference, was what you expected given the geopolitical context